



TRANSLATION HANKOOK

PORSCHE Inspection Catalog Chassis: Tires

Testing Procedure F.360.309

Rolling Circumference Difference FA/RA

Test Carrier:	Vehicle
Testing Facility:	Test Track
Status (Porsche):	19.01.2017
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(see target data sheet / "Zieldatenblatt")	

1 Purpose

This test method specifies a procedure for determining the rolling circumference difference between the front and rear axles on a vehicle.

The rolling circumference difference between the front and rear axles must be within a specified tolerance band for all-wheel drive vehicles, taking into account the axle ratios and under all operating conditions. The specified tolerance band is listed in the target data sheet.

2 Field of Application

This test method is valid for:

- Test tires
- Approval tests
- Retesting

3 Definition of Terms

The different axle ratios determine the zero line of the rolling circumference difference:

$$\frac{(i_{FA})}{(i_{RA})} = \frac{circumference_{FA}}{circumference_{RA}} \quad (Zero \ Line)$$

The limits of the rolling circumference differences are calculated as follows

$$Upper\ Limit = \frac{i_{FA}}{i_{RA}} + \frac{i_{FA}}{i_{RA}} * X_{UL} \geq \frac{circumference_{FA}}{circumference_{RA}}$$

$$Lower\ Limit = \frac{i_{FA}}{i_{RA}} + \frac{i_{FA}}{i_{RA}} * X_{LL} \leq \frac{circumference_{FA}}{circumference_{RA}}$$

with

i_{FA} = axle ratio front axle

i_{RA} = axle ratio rear axle

X_{UL} = upper limit in %

X_{LL} = lower limit in %

4 Responsibility

Test tires:

For each test tire specification, the rolling circumference difference shall be verified by the tire manufacturer at the time of result calculation. The results shall be made available to the PAG upon request.

Approval tests:

Testing for approval is performed and documented by the PAG. The results are made available to the tire manufacturer if required.

Retesting:

Retesting of approval tires can be arranged by the Porsche specialist department if required.

5 Description of the Testing Procedure

5.1 Measuring Equipment

5.1.1 Measured Variables / Assessment Variables

- Wheel rotational speeds at all wheels (pulses from ABS sensors)
- Vehicle speed

5.1.2 Installation of the sensors

N/A

5.1.3 Data Processing

N/A

5.2 Test Conditions

- Straight level roadway section of sufficient length
- Ambient temperatures for all-season/summer tires should be between 10° C and 34° C
- Ambient temperature for winter tires should be <15° C

5.3 Test Carrier / Test Set-Up

- Test vehicle in consultation with Porsche
- Loading condition: test vehicle incl. driver
- No new tires or tires from dry handling
- Rim width, diameter and contour at the rim seat of the test rim must correspond to the respective Original Porsche wheel (see target data sheet)
- Tire pressure: Vehicle air pressure in consultation with Porsche

5.4 Test Procedure

A step program is run in the speed levels 60, 150, 250km/h. All season and winter tires are tested up to the speed index. Further speed reductions of the third stage due to vehicle or route restrictions after consultation with Porsche.

The vehicle rolls freely straight ahead (clutch disengaged or at transmission stage "N") with steering angle 0° for the entire measurement period. The driving speed does not have to remain constant during the measurement period, but should correspond approximately to the speed steps on average. A measuring device adds up the respective ABS sensor pulses for each wheel separately during this measurement period. The measurements are performed three times per speed step.

6 Documentation

6.1 Data Analysis

Using the Excel evaluation sheet provided by Porsche. Make sure that the correct vehicle project is selected in the evaluation sheet.

6.2 Presentation of the results

In the form of the Porsche Excel evaluation sheet

6.3 Requirements

The tire must achieve the values specified in the target data sheet.

7 Applicable Documents

none

8 Distributor

EFF

Tire development partner

PES